Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item

Instructions

1. Select all the techniques that can enhance the privacy of data used in AI models. 1 / 1 point

- [x] Differential Privacy

  Nice work

  Correct! Differential privacy introduces noise to the data or queries, ensuring that individual entries remain indistinguishable while allowing accurate statistical analysis.

- [x] Federated Learning

  Nice work

  Correct! Federated learning facilitates the training of a global model across multiple devices while keeping data decentralized and private, enhancing overall privacy.

- [ ] Early Stopping
- [x] Homomorphic Encryption

  Nice work

  Correct! Homomorphic encryption enables computations on encrypted data without requiring decryption, thus preserving data confidentiality throughout processing.

- [ ] Data Normalization

2. Which of the following methods is most effective for protecting AI models from adversarial attacks? 1 / 1 point

- (•) Adversarial Training
- ( ) Batch Normalization
- ( ) Gradient Clipping
- ( ) Data Augmentation

  Nice work

  Correct! Adversarial training involves incorporating adversarial examples into the training process, enhancing the model's robustness against such attacks by explicitly teaching it to handle them.

3. What is the primary purpose of using Federated Learning in AI? 1 / 1 point

- ( ) To allow seamless integration of AI models into various applications
- (•) To enable multiple devices to collaboratively train a model without sharing raw data
- ( ) To reduce the computational load on a single device by distributing it
- ( ) To enhance the speed of training complex models

  Nice work

  Correct! Federated learning allows multiple devices to work together on model training while retaining their data locally, thereby ensuring data privacy and reducing data transmission risks.

4. In the context of reinforcement learning based adversarial attacks, what is the primary role of the reward signal? 1 / 1 point

- (•) To guide the learning process by providing feedback on actions
- ( ) To initialize the weights of the neural network
- ( ) To determine the structure of the neural network
- ( ) To provide labeled data for supervised learning

  Nice work

  Correct! The reward signal is crucial in reinforcement learning as it provides feedback to the agent about the quality of actions taken, guiding the learning and improvement process.

5. Which of the following best describes the primary difference between reinforcement learning based attacks and neural network based attacks? 1 / 1 point

- ( ) Neural network based attacks are more adaptive than reinforcement learning based attacks.
- (•) Reinforcement learning based attacks learn through trial and error while neural network based attacks rely on pre-trained models.
- ( ) Reinforcement learning based attacks are primarily used for image recognition while neural network based attacks are used for decision making.
- ( ) Reinforcement learning based attacks require labeled data while neural network based attacks do not.

  Nice work

  Correct! Reinforcement learning based attacks continuously adapt and improve through trial and error, whereas neural network based attacks utilize static models that have been pre-trained.

6. Select all statements that are true about reinforcement learning based adversarial attacks. 1 / 1 point

- [ ] They do not require a reward signal.
- [x] They aim to maximize a cumulative reward.

  Nice work

  Correct! The central goal of reinforcement learning is to maximize cumulative rewards over time, guiding the agent towards optimal behavior.

- [ ] They are primarily used in supervised learning scenarios.
- [x] They rely on a continuous feedback loop for learning.

  Nice work

  Correct! Reinforcement learning models depend on continuous interactions with the environment to refine their strategies and improve performance.

- [x] They can adapt to changes in the environment over time.

  Nice work

  Correct! Reinforcement learning models are designed to adapt and respond to evolving conditions in the environment, making them versatile and robust.

7. In the context of reinforcement learning, what is the primary role of the reward matrix? 1 / 1 point

- ( ) To store the Q values for each state-action pair
- ( ) To update the policy based on the observed rewards
- (•) To provide feedback to the agent based on the actions it takes
- ( ) To define the possible states and actions in the environment

  Nice work

  Correct! The reward matrix evaluates the agent's actions by providing feedback that informs the learning process and helps adjust the agent's behavior.

8. Which of the following are applications of reinforcement learning in AI and cybersecurity? (Select all that Apply.) 1 / 1 point

- [ ] Face recognition systems
- [ ] Speech-to-text systems
- [x] Automated trading systems

  Nice work

  Correct! Automated trading systems leverage reinforcement learning to optimize trading strategies and decision-making processes in financial markets.

- [x] Intrusion detection systems

  Nice work

  Correct! Reinforcement learning can be effectively used to enhance intrusion detection systems by improving their ability to identify and respond to security threats.

- [x] Adaptive traffic signal control

  Nice work

  Correct! Reinforcement learning is utilized to dynamically adjust traffic signal timings, optimizing traffic flow and reducing congestion in real-time.

9. Select all the components that are fundamental to reinforcement learning. 1 / 1 point

- [x] Q value

  Nice work

  Correct! The Q value is critical for evaluating the potential benefits of actions in specific states and guiding the agent's decision-making process.

- [x] Environment

  Nice work

  Correct! The environment provides the context and feedback necessary for the agent to learn and adapt its behavior.

- [ ] Gradient Descent
- [x] Agent

  Nice work

  Correct! The agent is a fundamental component of reinforcement learning, responsible for interacting with the environment and learning from it.

- [ ] Activation Function

10. Which of the following best describes the Q value in reinforcement learning? 1 / 1 point

- ( ) A matrix that stores the possible states and actions in the environment
- (•) A measure of the expected future rewards for an action taken in a given state
- ( ) A list of rules that defines how the agent should act at each state
- ( ) A mechanism to penalize the agent when a suboptimal action is taken

  Nice work

  Correct! The Q value represents the anticipated future rewards for taking a specific action in a particular state, guiding the agent's learning process.